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FOREST FIRE DETECTION SYSTEM USING MACHINE LEARNING



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Abstract

Forest fires are among the most destructive environmental disasters, causing severe ecological imbalance, biodiversity loss, atmospheric pollution, and economic damage. The increasing frequency of wildfires due to climate change, prolonged drought, and human negligence highlights the urgent need for early detection systems. Traditional fire detection methods such as watchtowers, manual patrols, and satellite monitoring often suffer from delayed detection, limited accessibility, and high operational costs. This paper presents an Internet of Things (IoT) and Machine Learning-based Forest Fire Detection System designed for real-time environmental monitoring and early warning. The system employs distributed sensors to measure temperature, humidity, and smoke levels. The collected data is processed using a microcontroller and analysed using intelligent threshold algorithms and machine learning models to improve detection accuracy. Upon identifying abnormal environmental patterns, the system automatically transmits alerts to concerned authorities through wireless communication modules. The proposed system emphasizes affordability, scalability, low power consumption, and real-time responsiveness. Experimental evaluation demonstrates improved detection speed and reduced false alarms compared to traditional approaches.

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